

Planning Report: Scalable Electronics for Modular Multi-Layer e-Skin

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Abstract

This planning report investigates scalable electronics for modular multi-layer e-skin, with particular attention to three system-level challenges: data scalability, power scalability, and calibration scalability. Building on the baseline platform developed by Devillard et al. and its later extension by Dutta et al., the report reviews the current literature, identifies key technical bottlenecks, and proposes preliminary experimental directions for a more compact and scalable modular architecture. As this is an early-stage planning report, the emphasis is placed primarily on literature review and theory-driven experimental design rather than on completed implementation. The experimental workflow presented here should therefore be understood as provisional and may be refined in response to results obtained at different stages of the project. Some figures were generated with AI assistance to help illustrate concepts and design ideas, and may not be fully accurate in scale, proportion, or geometry.

Word count: 3985

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1 Project Specification

Electronic skin (e-skin) is an important area in robotics because it enables distributed tactile sensing over large and curved surfaces, supporting safer interaction, richer perception, and more adaptive manipulation [1–5]. Recent work has advanced multi-modal sensing [6–13], large-area fabrication [10, 14–22], and robotic deployment [2, 4, 5, 9, 11, 12, 20, 21, 23–28]. However, much of this work still focuses on sensors [29–39], materials [12, 15, 16, 29, 40, 41], structures [9, 16], and AI-driven or bio-inspired algorithms [4, 5, 11, 12, 20, 21, 25–28, 42, 43], while giving less attention to a key barrier to practical deployment, namely **electronics scalability**.

A promising route is a **modular e-skin architecture**, where larger systems are built by integrating multiple modules without major redesign. Previous studies have already explored this direction [6–8, 10, 13, 18, 21, 40, 42, 44–46]. Building on the multi-modal, multi-layer platform of *Devillard et al.* [23] and its later extension by *Dutta et al.* [24], this project investigates how the baseline prototype can be redesigned into a compact and scalable modular system. The project is primarily hardware-oriented. It does not aim to replace the sensing principles of the baseline device, but to develop and evaluate an electronics architecture that supports miniaturisation, multi-module integration, and future multi-patch expansion while preserving useful tactile performance.

To address this aim, the project tests four hypotheses.

- **H1 (Core).** A miniaturised five-module patch can preserve the core sensing capability of the baseline platform, such that normal- and shear-force regression NMSE remain within 20% of the baseline under comparable experimental conditions.
- **H2 (Data scalability).** A hierarchical host–patch–module architecture can reduce communication load sufficiently to support scalable multi-module integration while keeping per-host bandwidth below a defined limit for several patches. In addition, event-driven algorithms can be combined with clock synchronization without violating H1.
- **H3 (Calibration scalability).** A repeatable calibration workflow can be established such that calibration transfer error after disassembly and reassembly remains below 10%, indicating that the system remains functional and maintainable as module count increases.
- **H4 (Power scalability).** A scalable patch-level power architecture can support stable operation of the miniaturised five-module patch, such that power consumption remains below a defined limit under continuous multimodal acquisition and scales approximately linearly with sampling rate and the number of enabled modules.

The main contribution of the project is therefore a miniaturised five-module scalable e-skin patch together with **a reference architecture for scalable data, power, and calibration management**. Larger goals, including multi-patch chaining, whole-arm deployment, and dual-arm robotic implementation, are treated as longer-term extensions rather than primary MSc deliverables [6, 19, 42, 47]. Success will be judged by whether the proposed architecture can manage communication load, preserve acceptable force-sensing accuracy relative to the baseline, and maintain calibration repeatability after reassembly.

2 Literature Review

This section begins with a detailed review of the two foundational papers on which this project is based. It then examines, summarises, and compares related studies in the field to identify the main bottlenecks and research gaps that constrain electronics scalability. Finally, it discusses a number of existing and potential approaches for addressing these limitations.

2.1 Overview

The development of electronic skin (e-skin) has progressed from individual tactile sensors and small arrays for basic contact or pressure detection [1, 2, 33, 38, 39, 43], to large-area flexible robotic skins that introduced challenges in wiring, readout, and deployment [18, 21, 44]. More recent work has moved towards bio-inspired multimodal systems integrating force, vibration, temperature, and texture sensing, supported by advances in soft materials [3, 26, 48]. As sensing scale and complexity have increased, AI, neuromorphic sensing, and event-driven processing have also become important for tactile inference and data handling [4, 27, 49]. Overall, e-skin development has shifted from simple tactile sensing to intelligent multimodal systems, with scalability and deployability becoming increasingly important [5, 50].

2.2 Baseline Prototype e-Skin and Its Extension Towards Embodied Tactile Perception

This project builds on the modular multi-layer e-skin platform introduced by *Devillard et al.* [23] and later extended by *Dutta et al.* [24]. Together, these studies provide the main hardware and methodological basis for the present work: the former establishes the prototype design and readout architecture, while the latter extends it towards embodied tactile perception.

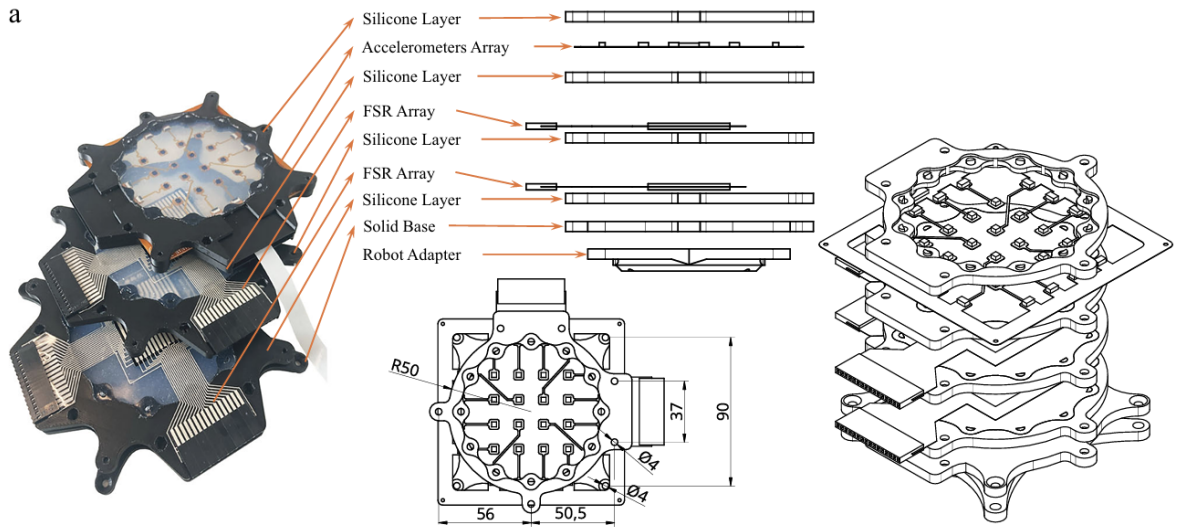


Figure 1: Structural design of the baseline modular multi-layer e-skin platform [24].

2.2.1 Modular Structure

As shown in **Figure 1**, the baseline e-skin is a modular¹, multi-modal artificial skin inspired by biological skin. It consists of multiple sensorized layers embedded within silicone, including an outer vibration-sensing layer, two internal force-sensing layers, silicone interlayers, a solid base, and a robot adapter. This layered design allows sensing configuration and local compliance to be varied while keeping the overall patch geometry consistent. In the later study, the same structure is further treated as an embodiment variable rather than a fixed sensing substrate.

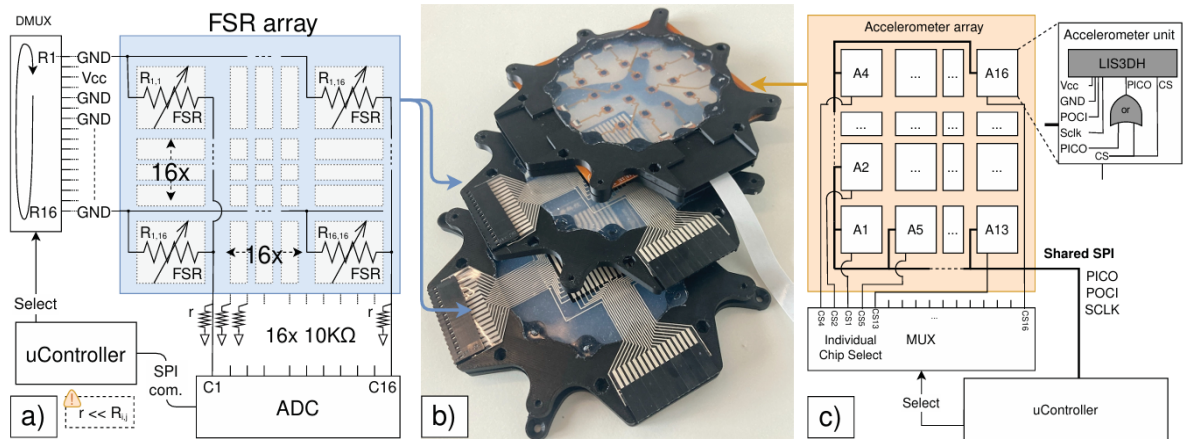


Figure 2: Baseline modular multi-layer e-skin platform and its sensing architecture [23]. (a) FSR array readout architecture. (b) Photograph of the assembled multi-layer e-skin prototype. (c) Accelerometer array architecture.

2.2.2 Hardware Structure

The prototype integrates force and vibration sensing within the same contact area. Two 16×16 *FSR arrays* provide 512 taxels for distributed force sensing, while a 4×4 array of *3-axis LIS3DH accelerometers* provides 48 vibration channels. The system uses *Ecoflex 00-31* and *DragonSkin 30A* as alternative silicone materials, with overall dimensions of approximately $120 \times 120 \times 25$ mm. As shown in **Figure 2(a)**, FSR readout uses matrix scanning with row–column addressing, while the accelerometer array uses SPI with multiplexed chip selection, as shown in **Figure 2(c)**. Data are acquired by a Teensy 4.1 at 700 Hz with real-time USB transmission [23]. A full summary of the hardware parameters is provided in **Appendix A Table A1**.

2.2.3 Algorithmic Implementation

The two papers differ mainly in algorithmic scope. *Devillard et al.* use task-specific models for force estimation and object-property prediction, whereas *Dutta et al.* introduce the **Latent Filter**, an action-conditioned deep state-space model for embodied tactile perception [24]. As shown in **Figure 3**, this model combines modality-specific encoders,

¹The term ‘**modular**’ in the paper [23] is different from the meaning of modular’ adopted in this report. Here, modular’ refers to local modularity within a single e-skin unit, namely a layered structure in which sensing and silicone layers can be detached, replaced, and reconfigured. It does not yet refer to a system-level modular network of multiple e-skin patches.

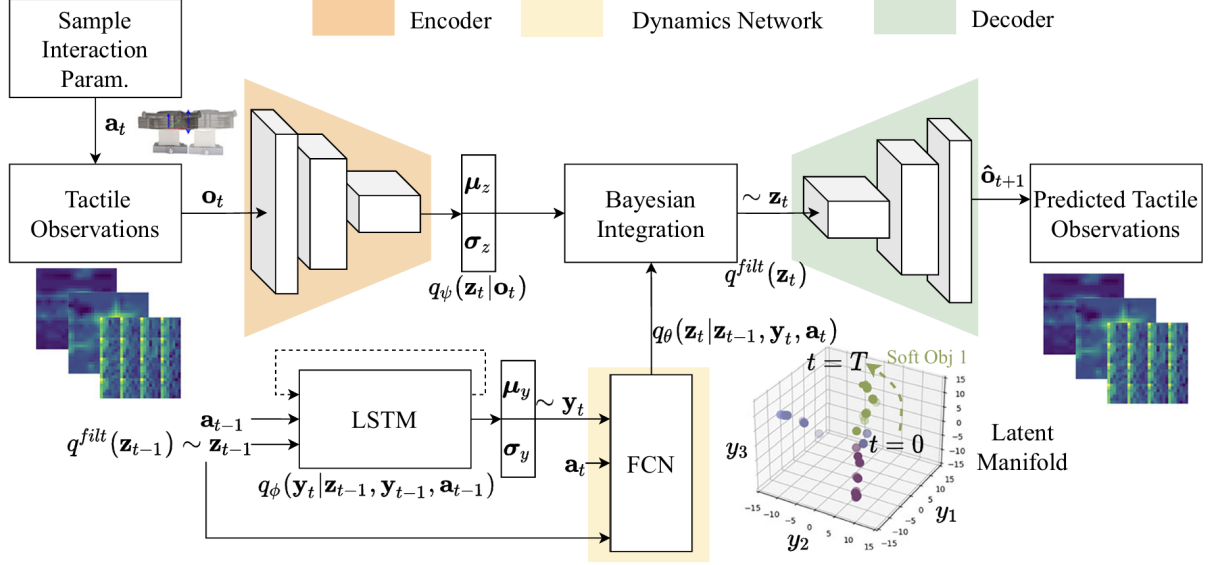


Figure 3: Architecture of the proposed Latent Filter for embodied tactile perception [24].

latent inference, dynamics modelling, and reconstruction. *The full Latent Filter algorithm is provided in **Appendix B Algorithm B1**.*

2.2.4 Experimental Design

Both studies use similar robotic experiments based on pressing, precession, and sliding over structured objects [23, 24]. The later work extends the earlier one by using a richer object set and incorporating robot proprioceptive and action information, while retaining the same multimodal sensing platform and 700 Hz acquisition setting.

2.2.5 Performance and Findings

Taken together, the two papers show that the platform can capture both quasi-static force and dynamic vibration signals in a robust and interpretable way. As shown in **Figure 4a**, the earlier work demonstrates reasonable normal- and shear-force estimation performance and shows that combining two FSR layers can improve force sensing [23]. As shown in **Figure 4b**, the later work further shows that multimodal sensing and interaction strategy strongly affect perceptual performance [24]. Overall, the baseline platform provides a strong starting point for this project while also exposing the need for a more compact and scalable electronics architecture.

Axis	Layers	Skin type	Normalized mean error (N)				Mean error (N)			
			Number samples				Number samples			
			5	10	20	40	5	10	20	40
X	Top	Ecoflex 00-31	0.035	0.032	0.029	0.029	0.489	0.448	0.410	0.408
		DragonSkin 30	0.036	0.031	0.030	0.027	0.504	0.437	0.429	0.386
	Bot	Ecoflex 00-31	0.040	0.037	0.034	0.032	0.551	0.518	0.469	0.445
		DragonSkin 30	0.042	0.040	0.035	0.032	0.595	0.558	0.501	0.449
	Top+Bot	Ecoflex 00-31	0.033	0.030	0.028	0.027	0.465	0.412	0.394	0.375
		DragonSkin 30	0.034	0.030	0.026	0.026	0.486	0.425	0.374	0.373
Y	Top	Ecoflex 00-31	0.028	0.027	0.023	0.023	0.481	0.464	0.401	0.392
		DragonSkin 30	0.035	0.031	0.030	0.028	0.481	0.427	0.414	0.383
	Bot	Ecoflex 00-31	0.032	0.030	0.029	0.027	0.548	0.520	0.495	0.459
		DragonSkin 30	0.041	0.039	0.035	0.033	0.568	0.542	0.478	0.449
	Top+Bot	Ecoflex 00-31	0.027	0.025	0.023	0.023	0.460	0.427	0.400	0.385
		DragonSkin 30	0.032	0.029	0.028	0.027	0.442	0.402	0.385	0.370
Z	Top	Ecoflex 00-31	0.030	0.028	0.024	0.025	2.280	2.136	1.876	1.898
		DragonSkin 30	0.028	0.024	0.024	0.023	2.235	1.960	1.915	1.827
	Bot	Ecoflex 00-31	0.033	0.032	0.030	0.029	2.544	2.460	2.285	2.212
		DragonSkin 30	0.032	0.031	0.028	0.026	2.604	2.460	2.233	2.092
	Top+Bot	Ecoflex 00-31	0.028	0.025	0.025	0.023	2.147	1.948	1.910	1.797
		DragonSkin 30	0.025	0.024	0.022	0.021	2.048	1.899	1.745	1.719

(a) Force estimation errors along the X, Y, and Z axes under different layer configurations, skin materials, and sample numbers.

Pressing			Precession			Sliding			
A_z (mm)	ω_z (Hz)	Traj. NMSE	$A_{\alpha,\beta}$ (mm)	$\omega_{\alpha,\beta}$ (Hz)	Traj. NMSE	$A_{x,y}$ (mm)	A_γ (°)	$\omega_{x,y,\gamma}$ (Hz)	Traj. NMSE
2.0	0.2	7.409	3.0	0.2	7.483	2.0	3.0	0.2	10.051
2.0	0.4	6.816	3.0	0.4	5.860	2.0	3.0	0.4	7.143
2.0	0.6	6.821	3.0	0.6	5.936	2.0	3.0	0.6	7.088
2.0	0.8	6.888	3.0	0.8	6.946	2.0	3.0	0.8	9.536
4.0	0.2	6.556	4.0	0.2	6.784	4.0	4.0	0.2	9.889
4.0	0.4	6.351	4.0	0.4	4.867	4.0	4.0	0.4	6.646
4.0	0.6	6.204	4.0	0.6	5.036	4.0	4.0	0.6	6.591
4.0	0.8	5.934	4.0	0.8	6.451	4.0	4.0	0.8	9.172
6.0	0.2	6.419	5.0	0.2	6.676	6.0	5.0	0.2	9.978
6.0	0.4	6.185	5.0	0.4	4.657	6.0	5.0	0.4	6.763
6.0	0.6	5.953	5.0	0.6	4.593	6.0	5.0	0.6	6.946
6.0	0.8	5.969	5.0	0.8	6.539	6.0	5.0	0.8	9.150
8.0	0.2	6.285	6.0	0.2	6.803	8.0	6.0	0.2	10.166
8.0	0.4	5.928	6.0	0.4	4.758	8.0	6.0	0.4	7.367
8.0	0.6	5.852	6.0	0.6	3.923	8.0	6.0	0.6	6.924
8.0	0.8	5.877	6.0	0.8	6.850	8.0	6.0	0.8	9.164

(b) Trajectory reconstruction NMSE under pressing, precession, and sliding motions with different amplitudes and frequencies.

Figure 4: Experimental results for motion trajectory reconstruction and force estimation.

2.3 Research Gap: No Scalable Reference Architecture for e-Skin Electronics

This project investigates how an existing e-skin prototype can be redesigned into a scalable, generalisable, and deployable modular platform, and is therefore positioned primarily as a hardware-oriented study. *Devillard et al.* provide the main hardware foundation, while *Dutta et al.* serve mainly as a software-side reference for later evaluation.

Although existing reviews discuss e-skin in terms of materials, sensing mechanisms, hardware, signal processing, and applications [3–5, 26, 43, 48], scalability is rarely treated as a systems-engineering problem. Many studies mention large-area deployment, but few define a modular power and data backbone that can scale from one patch to many [19, 21, 27, 51]. Similarly, modular e-skin studies mainly focus on function, structure, or materials rather than on data, power, and calibration scalability [6–8, 10, 13, 21, 40, 42, 44–46]. **Appendix C Table A3** and **Appendix C Table A2** summarise these limitations. The key gap is therefore the lack of a scalable reference architecture to support the transition from one patch to many.

2.4 Data Scalability

As e-skin systems scale, data volume increases rapidly, making bandwidth and synchronization central bottlenecks.

2.4.1 Bandwidth is Under-Specified

The baseline prototype already suggests that communication load can become a limiting factor [23]. Although multimodal sensing and high refresh rates improve tactile richness [6–13, 23, 24], they also greatly increase channel count and transmission demand. The key challenge is therefore to reduce communication load without significantly sacrificing sensing performance.

Hierarchical signal processing.

Hierarchical architectures such as *HEX-O-SKIN* and *ROBOSKIN* distribute processing locally before patch-level transmission [7, 8, 18, 21, 42, 46]. This improves maintainability and scalability, but existing implementations use lower sensor density and sampling rates than the present prototype, so hierarchical processing alone is unlikely to be sufficient.

Event-driven algorithms.

Event-driven methods reduce traffic by transmitting only salient changes [19, 27, 49, 52–54]. SoDP is effective but threshold-sensitive [19], while *Cao’s dual-layer design* is more relevant here because it better matches the two-layer FSR structure [52].

2.4.2 synchronization is under-specified

Data scalability also depends on reliable temporal alignment. Intra-module synchronization is required across sensing modalities, while inter-module synchronization is required across distributed controllers. Existing solutions include programmable units, synchronization protocols, and hardware-level implementations [55–57], but scalable multimodal e-skin still lacks a clear synchronization strategy.

2.5 Calibration Scalability

2.5.1 Calibration Does Not Scale with Sensor Count

As sensor count increases, calibration becomes a major bottleneck. Dense FSR arrays require per-taxel correction, while multi-layer systems also require cross-layer alignment for reliable shear-force inference. Because calibration depends on material properties, geometry, and assembly conditions, methods developed for one prototype may not transfer well to another. A scalable e-skin system therefore requires a calibration pipeline that is repeatable, transferable, and extensible.

Automatic *in situ* calibration.

Automatic *in situ* calibration improves deployment efficiency by calibrating sensors directly on the robot using controlled contact and force–torque sensing [58]. However, it remains essentially a taxel-level method.

Plenum-based array calibration.

Plenum-based calibration improves throughput by exposing an entire skin patch to a uniform pressure field, allowing many taxels to be calibrated in parallel [28]. This is more scalable than point-by-point calibration.

Limitations and possible improvements.

Neither approach fully addresses dense modular e-skin. The *in situ* method remains limited by hysteresis and coupling effects [58], while the plenum-based method is constrained by pressure noise, leakage, and chamber robustness [28]. Current methods therefore do not yet provide a complete solution for high-density, multi-layer, modular calibration.

2.6 Power Scalability

2.6.1 Power and Thermal Constraints Are Not Treated as First-Class Design Variables

As e-skin systems scale in area, sensor density, and functional complexity, power consumption becomes a systems-level constraint. This is especially important for large or mobile robotic platforms, where additional sensors, controllers, and communication interfaces increase both electrical demand and thermal load, with possible effects on sensing stability and long-term reliability.

However, power and thermal issues are still rarely treated as core design variables. Existing work has proposed partial solutions, including event-driven acquisition [19, 27, 53], structural simplification [9], neuromorphic approaches [25], and modular patch-level integration [13, 21, 23]. These studies address parts of the problem, but do not define a complete power architecture for modular multi-layer e-skin.

Distributed energy perspectives for scalable robotics.

Mukherjee et al. argue that energy in robotics should be treated as a system-architecture problem rather than only a battery-capacity issue [59]. This perspective is relevant here because scalable e-skin depends not only on supplying power, but also on how it is distributed and regulated.

Limitations and possible improvements.

The literature still lacks a general reference architecture for modular multi-layer e-skin power design, particularly for power-tree design, domain isolation, subsystem monitoring, and power scaling with system size. This is the gap targeted by the present project.

3 Implementation Plan and Objectives

This section provides a detailed description of the final objectives of the project and the milestones for each phase.

3.1 Primary Objective

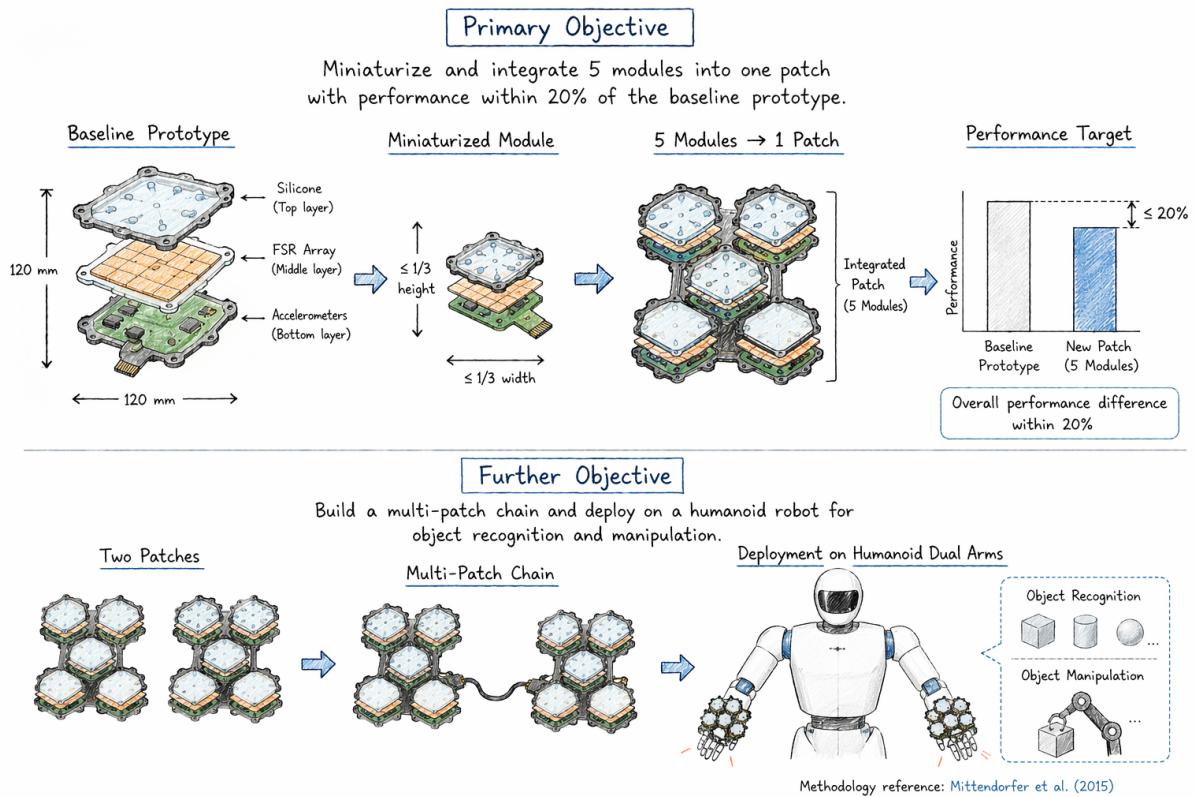


Figure 5: Figure illustration of the project objectives.

The primary objective of this project is to transform the current modular multi-layer e-skin prototype into a more compact and scalable system. Specifically, the electronics and structural footprint will be reduced to at least one third of the original size while preserving the core sensing functions of the baseline design.

Building on this miniaturised design, the project aims to integrate five e-skin modules into a single patch and evaluate it using a protocol similar to that in Section 2.2.4 [23]. The target is to keep tactile sensing performance close to the baseline prototype, defined here as the *Ecoflex 00-31* structure with both sensing layers. Using **Figure 4a** as reference, the final mean force estimation error should remain within 20% of baseline:

- **X-axis shear force:** 0.375 ± 0.075 N

- **Y-axis shear force:** 0.385 ± 0.077 N
- **Z-axis normal force:** 1.797 ± 0.3594 N

Force estimation error is used as the main criterion because it more directly reflects core tactile sensing capability than trajectory RMSE shown in *Figure 4b*. *Figure 5* illustrates the objective more clearly.

3.2 Intermediate Milestones

To achieve the final objective, the project is divided into five intermediate milestones, as illustrated in *Figure 6*.

O0. Modular system architecture (preliminarily completed in theory)

Define the overall scalable system architecture:

- **O0.1** Define module-level and patch-level interfaces for sensing, communication, and power.
- **O0.2** Establish an integration strategy for combining five miniaturised modules into one patch.
- **O0.3** Specify the partitioning between module electronics, patch coordination, and host-side processing.
- **O0.4** Ensure extensibility towards future multi-patch deployment.

O1. Data scalability

Establish reliable multimodal data acquisition for the five-module patch:

- **O1.1** Build a continuous streaming pipeline for the integrated patch.
- **O1.2** Maintain low frame drop, low timing jitter, and stable packet delivery.
- **O1.3** Preserve temporal alignment between FSR and accelerometer signals.
- **O1.4** Assess host-side logging and processing load for future expansion.

O2. Power scalability

Evaluate whether the integrated patch remains energetically practical:

- **O2.1** Develop a power architecture for stable five-module operation.
- **O2.2** Quantify current draw as a function of sampling rate. ²
- **O2.3** Quantify current draw as a function of enabled modules or sensors.
- **O2.4** Assess suitability for future dual-patch deployment.

²Preliminary Power Budget calculate in *subsection 7.3*

O3. Calibration scalability

Develop a repeatable and transferable calibration workflow:

- **O3.1** Define a repeatable calibration procedure for the integrated patch.
- **O3.2** Define a synchronization calibration procedure.
- **O3.3** Control inter-module variation across the assembled patch.
- **O3.4** Verify post-reassembly performance against the baseline.

O4. Reference design package

Deliver a reusable package for future e-skin development:

- **O4.1** Consolidate the electronics architecture, firmware, and calibration workflow.
- **O4.2** Document module design, patch integration, and validation in reproducible form.
- **O4.3** Summarise validation results for future comparison.
- **O4.4** Deliver a reference toolkit for future extensions.

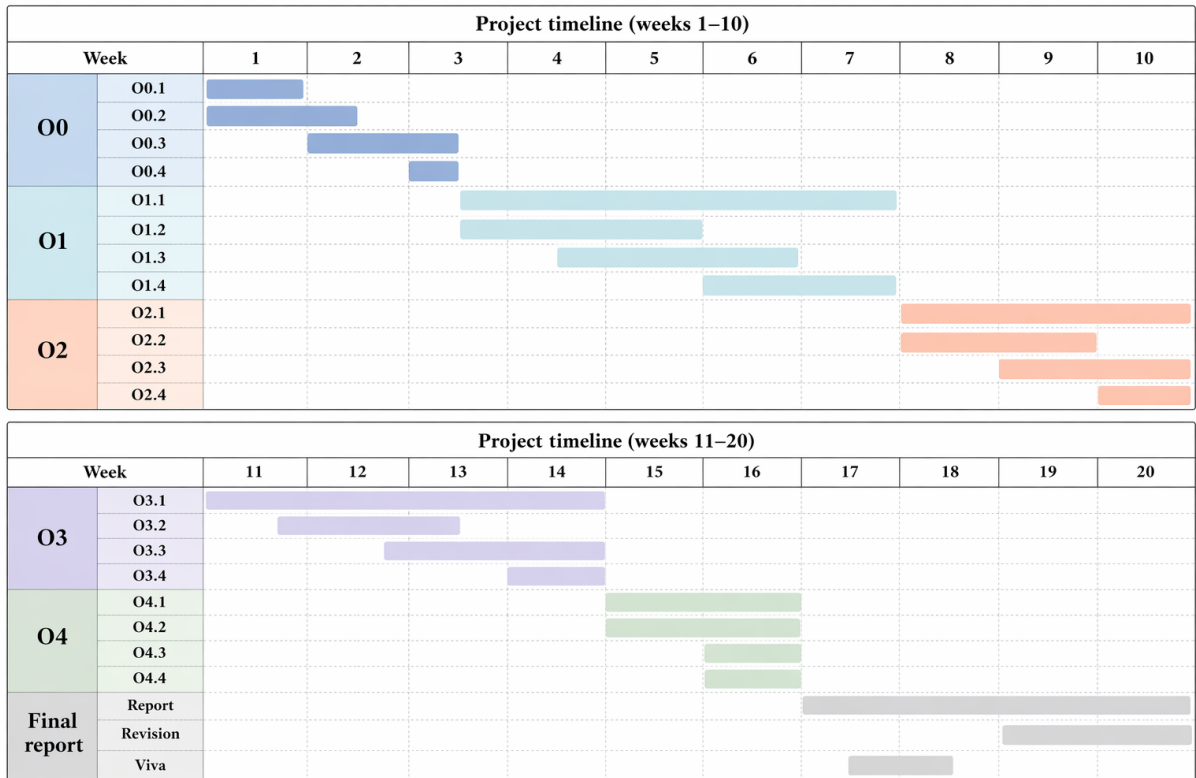


Figure 6: Project timeline over 20 weeks. The first phase, covering Weeks 1–10, focuses on modular system architecture, data scalability, and power scalability. The second phase, covering Weeks 11–20, focuses on calibration scalability, delivery of the reference design package, final report writing, revision, and viva preparation.

4 Risks and Mitigations

As the project progresses from a single baseline prototype to a miniaturised five-module patch and, potentially, a multi-patch system, several technical risks may affect system reliability and scalability. The main risks concern high-throughput streaming, timing consistency, analog sensing quality, calibration transferability, and future system expansion. **Table A5** summarises the key risks, causes, and mitigation strategies.

At the current stage, data instability and timing inconsistency are the most critical because they determine whether the five-module patch can operate as a coherent multimodal sensing system. Calibration-related risks become more important after assembly and re-assembly, when small mechanical variations may reduce repeatability. Multi-patch scaling is treated as a forward-looking risk so that early design choices do not constrain future expansion.

5 Evaluation

The evaluation plan is organised around four main milestones. Table A6 defines the metrics for **O1 data scalability**, including streaming reliability, packet stability, inter-modal temporal alignment, and host-side readiness for future expansion. Table A7 addresses **O2 power scalability**, covering stable power delivery, current draw versus sampling rate, current draw versus enabled modules, and the suitability of the resulting power budget for future dual-patch deployment. Table A8 evaluates **O3 calibration scalability**, including calibration repeatability, synchronization calibration, inter-module consistency, and post-reassembly performance retention. Table A9 defines the criteria for **O4 reference design package**, focusing on completeness, reproducibility, validation coverage, and future usability.

Together, these tables translate the project objectives into measurable criteria by specifying metrics, protocols, acceptance thresholds, and statistical analysis where appropriate. In addition, preliminary outcomes from **O0**, discussed in **Section 7.1.2**, are summarised in **Table 1** and **Table 2**, which provide an initial basis for the current architecture choices.

6 Ethical Analysis

This project involves relatively low direct ethical risk because it does not use human participants, animal subjects, or biological specimens. The work is based on robotic e-skin hardware and controlled robot–object interactions, and the collected data are non-personal sensor and calibration data. The main ethical considerations therefore concern safe engineering practice, responsible data handling, and the longer-term implications of tactile sensing in robotics.

A project-specific issue is safety during robotic testing and waste generated through iterative prototyping. Since experiments might will be conducted on a *Kinova robotic arm*, testing must follow laboratory risk assessments and established robot-safety procedures, including collision prevention, emergency stop access, workspace clearance, and supervised operation. More broadly, the work should remain consistent with recognised standards such as *ISO 10218* and *ISO/TS 15066* for future human–robot interaction contexts. In

addition, PCB fabrication, silicone moulding, and component replacement may generate material waste, so unnecessary iteration should be minimised and waste should be handled responsibly.

Overall, the ethical risk is low, but safe experimentation and responsible design remain essential.

7 Preliminary Results

At present, the work completed in this project is mainly theoretical. In particular, the main progress so far has been the development of the Data Throughput Model and the design of the Modular System Architecture.

7.1 Modular System Architecture

7.1.1 Module Structure

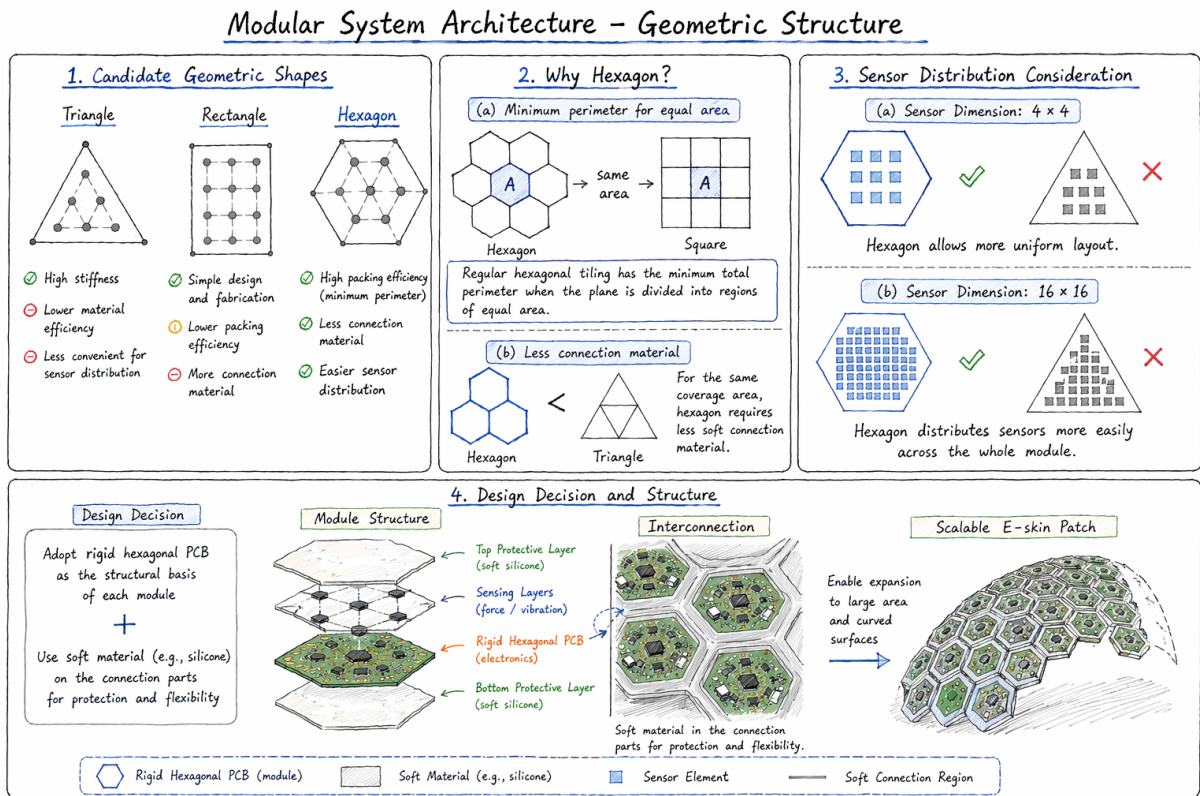


Figure 7: Figure illustration of the proposed geometric structure in the modular e-skin architecture.

At present, the regular geometric forms most commonly adopted in modular e-skin designs are triangular [7, 8, 42], rectangular [6], and hexagonal [7, 8, 13, 42] units. These geometries are widely used because they provide clear tessellation rules, facilitate physical assembly, and support scalable integration over larger sensing areas.

Currently, some studies have compared triangular and hexagonal mechanical structures and found that triangular configurations can provide higher bending and torsional stiffness

[60]. However, these advantages are not the primary concern for e-skin scalability. By contrast, a well-known study from a biomimetic perspective explains why honeycombs are predominantly hexagonal: when a plane is divided into regions of equal area, regular hexagonal tiling achieves the minimum total perimeter [61].

In other words, for the same coverage area, hexagonal modules require less compliant interconnection material between adjacent units. In addition, given that the sensor dimensions in this project are 4×4 and 16×16 , a regular hexagon provides a more practical basis than a triangle for distributing sensors across the entire module area.

For these reasons, the proposed design will preliminarily adopt a **rigid hexagonal PCB** as the structural basis of each module, while using compliant materials such as silicone to protect the interconnecting regions. This configuration is expected to better support future e-skin expansion over large areas and curved surfaces. **Figure 7** illustrates modular structure more clearly.

7.1.2 Macro Structure

The system hierarchy is defined as follows:

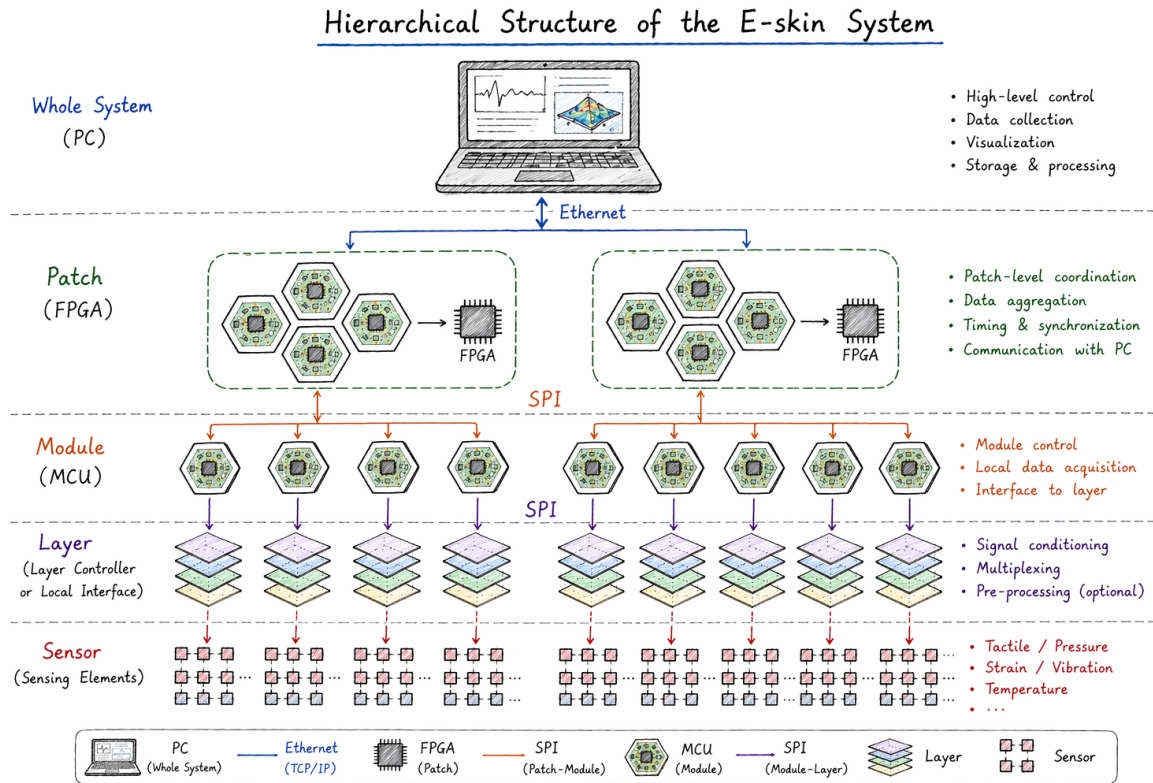


Figure 8: Figure illustration of the hierarchical structure of the proposed modular e-skin system.

This structure is intended to support both modular integration and future scalability.

Hierarchical Structure

At the macro level, the proposed e-skin follows a hierarchy from system to patch, module,

layer, and sensor. The *whole system* is currently a PC responsible for high-level control, logging, and processing. Each *patch* contains a patch-level controller that aggregates data from five modules, aligns timing, packetizes the data stream, and communicates with the host via Ethernet. Each *module* is controlled by a local MCU, and is further divided into sensing *layers* and individual *sensors*.

A key design question is whether the patch-level controller should be a high-performance MCU or an FPGA. Although an FPGA increases cost and development complexity, it is better suited to parallel aggregation, deterministic timing, synchronization, and SPI-to-Ethernet bridging as system scale increases. For this reason, the current design provisionally adopts an FPGA at the patch level. **Table 1** presents the preliminary decision matrix, and **Figure 8** illustrates the hierarchy.

Table 1: Preliminary decision matrix for the patch-level controller selection. Higher scores indicate a more favorable option.

Criterion	Weight	High-performance MCU	FPGA
Hardware cost	0.20	5	2
Bring-up simplicity / development risk	0.20	5	2
Parallel multi-module aggregation	0.20	2	5
Synchronization	0.20	3	5
SPI-to-Ethernet bridging scalability	0.20	2	5
Weighted total	1.00	3.4	3.8

Communication Protocol

The communication protocol follows the same hierarchy. Communication between the PC and each patch uses **Ethernet**, which provides a robust host-to-patch link and is supported by the DTM analysis in Section 7.2.3. Within each patch, **SPI** is provisionally retained between the patch-level controller and the five modules. Although SPI is not ideal for long-term scaling, it remains suitable at this stage because distances are short, module count is limited, and deterministic high-throughput communication is required. Alternative protocols such as *RS-485*, *CAN-FD*, and *LVDS* involve either lower effective throughput or greater physical-layer complexity; a comparison is given in **Table A4**. SPI is therefore used here as a practical short-range solution.

A second open question concerns the module-level controller. Reusing one Teensy 4.1 per module is likely to be inefficient, so two alternatives are considered: a *master-slave* design and a *lower-cost MCU* design. The latter currently appears more favorable in cost, integration, and scalability. **Table 2** summarises this preliminary comparison; the final choice will be made after O0.2.

Table 2: Decision matrix for module-level controller selection. Higher scores indicate a more favorable option.

Criterion	Weight	Master–slave	Cheaper MCU
Hardware cost	0.35	4	5
Wiring / integration simplicity	0.20	2	4
Determinism	0.20	3	4
Firmware reuse	0.15	5	3
Scalability / modularity	0.10	2	5
Weighted total	1.00	3.35	4.35

7.1.3 Edge Computing (Optional)

Edge computing is considered here as an optional extension of the proposed architecture. The core idea is to distribute part of the computation across different levels of the system, rather than transmitting all raw data to the final host controller. This concept has already been explored in recent e-skin research designed by *Li et al.*,2026 [53].

In the present design, a similar idea could be introduced through the hierarchical structure by adding encoder and decoder functions at different levels. In principle, each level could first process its own data locally, for example through feature extraction or compression, and then transmit only the reduced representation to the upper or lower level, where further decoding or interpretation would be performed. A possible future direction is to explore a VAE-based method for this purpose. However, this remains only a preliminary concept at the current stage. The specific implementation strategy will require further study and practical validation.

7.2 Data Throughput Model

This section estimates the raw data throughput at each level of the hierarchy.

7.2.1 DTM for each module

Each module contains two FSR layers and one accelerometer array, giving

$$C_{\text{module}} = 512 + 48 = 560$$

channels per frame. At 700 Hz, the sample throughput is

$$R_{\text{samples,module}} = 560 \times 700 = 392,000 \text{ samples/s}$$

Assuming 16-bit encoding³, the raw module bandwidth is

$$R_{\text{module,raw}} = 392,000 \times 2 = 784,000 \text{ B/s} \approx 0.784 \text{ MB/s}$$

With 20 B/frame metadata (timestamp, frame counter, module ID, flags, and CRC),

³In this section, B denotes byte and b denotes bit. 1 MB $\approx$ 1,000 kB $\approx$ 1,000,000 B $\approx$ 8,000,000 b.

$$H_{\text{meta}} = 20 \text{ B/frame}$$

the effective frame size is

$$D_{\text{module}} = 1120 + 20 = 1140 \text{ B/frame}$$

and the total module bandwidth becomes

$$R_{\text{module,total}} = 1140 \times 700 = 798,000 \text{ B/s} \approx 0.8 \text{ MB/s} \approx 6.4 \text{ Mb/s}$$

7.2.2 DTM for each patch

One patch contains five modules, so the aggregated payload is

$$D_{\text{patch,agg}} = 5 \times 1140 = 5700 \text{ B/frame}$$

Adding 20 B patch-level metadata gives

$$D_{\text{patch,total}} = 5700 + 20 = 5720 \text{ B/frame}$$

and the FPGA-side output bandwidth is

$$R_{\text{patch,FPGA}} = 5720 \times 700 = 4,004,000 \text{ B/s} \approx 4.0 \text{ MB/s} \approx 32.0 \text{ Mb/s}$$

Since this exceeds the conventional Ethernet MTU, fragmentation is required. With a maximum UDP payload of 1472 B,

$$N_{\text{pkt}} = \left\lceil \frac{5720}{1472} \right\rceil = 4$$

Assuming 66 B Ethernet overhead per packet,

$$D_{\text{patch,eth}} = 5720 + 4 \times 66 = 5984 \text{ B/frame}$$

so the required Ethernet bandwidth per patch is

$$R_{\text{patch,eth}} = 5984 \times 700 = 4,188,800 \text{ B/s} \approx 4.19 \text{ MB/s} \approx 33.5 \text{ Mb/s}$$

7.2.3 DTM for the whole system and link budget

For two identical patches,

$$R_{\text{total,eth}} = 2 \times 4,188,800 = 8,377,600 \text{ B/s} \approx 8.38 \text{ MB/s} \approx 67.0 \text{ Mb/s}$$

This corresponds to

$$U_{100} = \frac{8.38}{12.5} \approx 67.0\%$$

for **100BASE-TX**, and

$$U_{1000} = \frac{8.38}{125} \approx 6.7\%$$

for **1000BASE-T**. The remaining headroom is

$$H_{100} = 12.5 - 8.38 = 4.12 \text{ MB/s}$$

$$H_{1000} = 125 - 8.38 = 116.62 \text{ MB/s}$$

with theoretical saturation at

$$N_{\text{max},100} = \frac{12.5}{4.19} \approx 2\text{--}3 \text{ patches}$$

for **100BASE-TX**, and

$$N_{\text{max},1000} = \frac{125}{4.19} \approx 29 \text{ patches}$$

for **1000BASE-T**.

At the current stage, **100BASE-TX** is sufficient for two patches, whereas larger-scale applications will likely require **1000BASE-T**. The initial implementation therefore adopts **100BASE-TX**.

7.3 Preliminary Power Budget

Although O2.2 and O2.3 will provide direct current measurements in later stages, a rough estimate is useful at this point. For the accelerometer array, the LIS3DH datasheet reports a typical current of about 11–20 μA . For 16 sensors per module and 5 modules per patch, this gives a total accelerometer current of approximately

$$I_{\text{ACC,total}} \approx 0.88\text{--}1.60 \text{ mA}$$

or

$$P_{\text{ACC,total}} \approx 2.9\text{--}5.3 \text{ mW}$$

assuming a 3.3 V supply.

By contrast, controller-side power is likely to dominate. A Teensy 4.1 consumes approximately 100 mA at 600 MHz, so retaining one MCU on each of the five modules would require roughly

$$I_{\text{MCU,total}} \approx 500 \text{ mA}$$

or

$$P_{\text{MCU,total}} \approx 1.65 \text{ W}.$$

FSR divider losses are more dependent on contact conditions, but under a simple 3.3 V voltage-divider assumption, one actively loaded taxel may already dissipate power on the order of sub-mW to about 1 mW. This suggests that aggregate FSR losses may also become significant when many taxels are simultaneously engaged.

These values are only preliminary estimates, but they already indicate that controller-side power is likely to dominate over accelerometer power, while FSR readout losses may become relevant under dense contact conditions. The estimates will be refined through direct measurements in O2.2 and O2.3.

Appendix

A. Additional Hardware Information

Table A1: Summary of key hardware parameters of the baseline modular multi-layer e-skin platform.

Parameter	Value	Notes
Prototype size	$120 \times 120 \times 25$ mm	Overall e-skin dimensions.
Silicone materials	Ecoflex 00-31, DragonSkin 30A	Soft and relatively stiff embodiments.
FSR layers	$2 \times 16 \times 16$ arrays	512 taxels in total.
FSR cell size	2×2 mm ²	With 1 mm pitch.
Accelerometer layer	4×4 LIS3DH array	16 sensors and 48 channels in total.
Accelerometer spacing	15 mm	Inter-sensor spacing.
LIS3DH size	3×3 mm ²	Sensor footprint.
LIS3DH sensitivity	0.98 mg/LSB	Reported in the prototype paper.
LIS3DH range	$\pm 16g$	Dynamic range.
Microcontroller	Teensy 4.1	Main acquisition unit.
Clock speed	600 MHz	Supports synchronized high-rate sampling.
Sampling rate	700 Hz	For FSR and accelerometer acquisition.
Communication	USB HS, 480 Mbit/s	Real-time host streaming.
FSR readout	Demux + 16-channel ADC	Matrix-scanned architecture.
Accelerometer bus	SPI	Selected over I ² C for speed.

B. Latent Filter Algorithm Pseudocode

Algorithm B1 Training logic of the action-conditioned Latent Filter

Require: tactile observations $o_{1:T}$, action observations $a_{1:T}$, parameters θ, ψ, ϕ

Require: encoder $q_\psi(z_t | o_t)$, inference model $q_\phi(y_t | z_{t-1}, y_{t-1}, a_{t-1})$

Require: dynamics model $p_\theta(z_t | z_{t-1}, y_t, a_t)$, decoder $p_\theta(o_t | z_t)$

Ensure: trained latent filter parameters

```

1: Initialize  $\theta, \psi, \phi$ 
2: for each training epoch do
3:   for each sequence  $(o_{1:T}, a_{1:T})$  do
4:     Initialize  $z_0, y_0$ 
5:      $\mathcal{L} \leftarrow 0$ 
6:     for  $t = 1$  to  $T$  do
7:        $(\mu_{z_t}, \sigma_{z_t}) \leftarrow q_\psi(z_t | o_t)$  ▷ encode tactile observation
8:       if  $t > 1$  then
9:          $(\mu_{y_t}, \sigma_{y_t}) \leftarrow q_\phi(y_t | z_{t-1}, y_{t-1}, a_{t-1})$  ▷ infer hidden latent state
10:         $p_\theta(z_t | z_{t-1}, y_t, a_t)$  ▷ predict latent prior
11:         $q^{\text{filt}}(z_t) \leftarrow \text{BayesFuse}(q_\psi(z_t | o_t), p_\theta(z_t | z_{t-1}, y_t, a_t))$  ▷ Bayesian integration
12:      else
13:         $q^{\text{filt}}(z_t) \leftarrow q_\psi(z_t | o_t)$  ▷ use encoder output directly
14:      end if
15:       $z_t \sim q^{\text{filt}}(z_t)$  ▷ sample filtered latent variable
16:       $\hat{o}_t \sim p_\theta(o_t | z_t)$  ▷ reconstruct tactile observation
17:       $\mathcal{L}_{\text{rec}} \leftarrow -\log p_\theta(o_t | z_t)$  ▷ reconstruction loss
18:      if  $t > 1$  then
19:         $\mathcal{L}_z \leftarrow \text{KL}(q^{\text{filt}}(z_t) \| p_\theta(z_t | z_{t-1}, y_t, a_t))$  ▷ latent dynamics regularization
20:         $\mathcal{L}_y \leftarrow \text{KL}(q_\phi(y_t | \cdot) \| p(y_t | a_t, N))$  ▷ hierarchical prior regularization
21:      else
22:         $\mathcal{L}_z \leftarrow 0, \mathcal{L}_y \leftarrow 0$ 
23:      end if
24:       $\mathcal{L} \leftarrow \mathcal{L} + \mathcal{L}_{\text{rec}} + \mathcal{L}_z + \mathcal{L}_y$ 
25:    end for
26:    Update  $\theta, \psi, \phi$  using Adam and backpropagation
27:  end for
28: end for

```

C. Comparison between different modular e-skin prototype

Table A2: Comparison of representative modular e-skin systems from the perspective of scalability

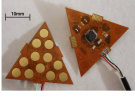
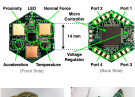
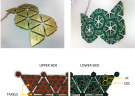
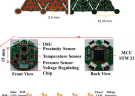
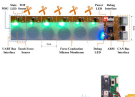
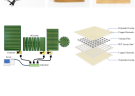
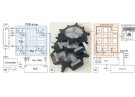
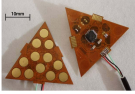
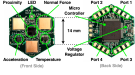
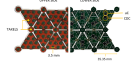
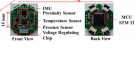
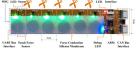
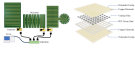
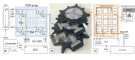
Image	Prototype	Data scalability	Power scalability	Calibration scalability
	<i>ROBOSKIN Ohmura 2006</i> [44]	Curved-surface scalability	N/A	N/A
	<i>Cannata 2008</i> [46]	Embedded humanoid skin	Low-power optical design	N/A
	<i>Mittendorfer 2011</i> [42]	Local preprocessing	Active tactile modules	N/A
	<i>Maiolino 2013</i> [21]	Large-area patch tiling	N/A	Thermal-drift compensation
	<i>ProxySKIN 2024</i> [13]	EtherCAT-based distributed sensing	N/A	N/A
	<i>TacSuit 2024</i> [10]	Hierarchical event fusion	Local computing MCU	N/A
	<i>Xue 2025</i> [6]	Adjustable module size	N/A	N/A
	<i>Hierarchical Safety Framework 2024</i> [62]	Whole-body sensing	N/A	N/A
	<i>Living Layers 2025</i> [40]	Layered modular construction	N/A	N/A
	<i>Full-body Modular E-skin 2025</i> [45]	Full-body deployment	N/A	N/A
	<i>Modular Multi-Layer e-Skin 2025</i> [23]	Prototype-level multimodal DAQ	Prototype-level only	Prototype-level only

Table A3: Comparison of representative modular e-skin systems

Image	Prototype	Module form	Multilayer	Function	Sensors	Main characteristics
	<i>ROBOSKIN Ohmura 2006</i> [44]	Conformal patch	No	Curved-surface touch; conformal coverage	Pressure-sensitive tactile elements	Conformal scalable skin for curved robot surfaces
	<i>Cannata 2008</i> [46]	Triangular cell	No	Whole-body touch; contact detection	Embedded artificial skin sensing elements	Early embedded humanoid skin
	<i>Mittendorfer 2011</i> [42]	Hexagonal cell	No	Contact detection; multimodal tactile sensing	Multimodal tactile sensing module	Local preprocessing
	<i>Maiolino 2013</i> [21]	Triangular cell	No	Normal pressure; large-area touch	Capacitive tactile taxels	Robust patch tiling
	<i>ProxySKIN 2024</i> [13]	Triangular cell	No	Contact sensing; pre-contact detection; HRI	Tactile sensors + ToF proximity sensors	Multi-modal HRI skin
	<i>TacSuit 2024</i> [10]	Hexagonal cell	No	Pressure; proximity; vibration; temperature; collision detection	Pressure sensor, proximity sensor, accelerometer, temperature sensor	Hierarchical event fusion
	<i>Xue 2025</i> [6]	Rectangular cell	No	Collision detection; robot body touch	Multimodal flexible e-skin sensors	Adjustable module size
	<i>Hierarchical Safety Framework 2024</i> [62]	Irregular patch	No	Whole-body safety monitoring; contact-aware robot safety	Whole-body tactile sensors	Hierarchical safety framework
	<i>Living Layers 2025</i> [40]	Layered patch	Yes	Modular e-skin construction	Bacterial-cellulose-based modular e-skin platform	Biofabricated layering
	<i>Full-body Modular E-skin 2025</i> [45]	Irregular patch	No	Whole-body touch; tactile gesture recognition	Pressure sensing points + pose information	Geometry-aware full body
	<i>Modular Multi-Layer e-Skin 2025</i> [23]	Multi-layer patch	Yes	Normal pressure; vibration; embodied tactile perception	FSR array + accelerometer layer	Core project platform

D. Summary of differnet protocols

Table A4: Comparison of mainstream transmission protocols

Protocol	Bandwidth	Max data rate	Signal wires	Typical topology / distance	Key features
I ² C	Low to medium	100 kb/s, 400 kb/s, 1 Mb/s, 3.4 Mb/s	2	Short-distance, multi-drop bus	Low pin count; easy expansion; low speed
I3C	Medium	12.5 Mb/s SDR	2	Short-distance sensor bus	Higher speed; backward-compatible; efficient bus management
SPI	Medium to high	Typically up to tens of Mb/s	4	Short-distance point-to-point or multi-slave	High speed; full duplex; simple protocol
UART	Low to medium	Typically kb/s to several Mb/s	2	Point-to-point, short to medium distance	Simple interface; no clock; low complexity
CAN FD	Medium	1 Mb/s arbitration, up to 8 Mb/s data phase	2	Differential multi-drop bus	Robust; anti-interference; good error detection
RS-485	Medium	Up to 10 Mb/s at short distance	2 or 4	Long-distance differential bus	Long range; strong noise immunity; industrial use
USB 2.0	High	1.5 Mb/s, 12 Mb/s, 480 Mb/s	2	Host-device, short cable	Plug-and-play; mature ecosystem; moderate complexity
USB 3.2	Very high	5 Gb/s, 10 Gb/s, 20 Gb/s	Depends on connector	Host-device, short cable	Very high throughput; complex PHY
100BASE-TX Ethernet	High	100 Mb/s	4	Switched network, up to 100 m	Mature standard; reliable; network capable
1000BASE-T Ethernet	Very high	1 Gb/s	8	Switched network, up to 100 m	High throughput; scalable; complex hardware

E. Risk Metric

Table A5: Main project risks and mitigation strategies. Likelihood and impact are scored on a 1–5 scale, and the risk score is calculated as $L \times I$.

Risk	Why it happens	$L \times I$	Mitigation
Data dropouts	bandwidth spikes, host buffering, or framing errors during continuous acquisition	$3 \times 4 = 12$	use fixed-size frames, CRC, ring buffers, and stress-test logging before full integration
Timing jitter	non-deterministic firmware scheduling and asynchronous acquisition paths	$3 \times 4 = 12$	use timer-driven acquisition, DMA where possible, and explicit timestamp validation
Noisy force readings	ADC reference instability, ground bounce, and analog–digital interference	$3 \times 3 = 9$	separate analog and digital domains, improve grounding, and add filtering and shielding if needed
Calibration does not transfer	silicone assembly variance, layer shift, and reassembly inconsistency	$4 \times 4 = 16$	log full calibration metadata, add an alignment step, and evaluate repeatability after reassembly
Multi-patch scaling fails	bus contention, synchronization drift, and increased communication overhead	$3 \times 5 = 15$	use shared synchronization, deterministic framing, and validate bandwidth limits early with DTM
PCB bring-up failure or fabrication delay	miniaturisation may require more than one board spin, while fabrication and assembly delays can accumulate	$4 \times 5 = 20$	plan for at least two board spins, place the second order before full debugging of the first, and keep a fallback test board if possible
Silicone mould or sensor-layer misalignment	shrinking the footprint increases assembly sensitivity and makes layer registration more difficult	$4 \times 4 = 16$	introduce alignment features in the mould, use assembly jigs, and verify registration before final bonding
FPGA toolchain ramp-up	FPGA development adds learning overhead, bring-up risk, and debugging complexity compared with MCU-only designs	$3 \times 4 = 12$	start toolchain learning early, validate a minimal FPGA data path first, and keep the MCU option as a fallback

E. Evaluation Metrics

Table A6: Evaluation metrics for O1 data scalability.

Obj.	Metric	Protocol	Acceptance criterion	Statistical test
O1.1	Streaming success rate	10 repeated multimodal streaming runs, each lasting 10 min, from the integrated five-module patch at 700 Hz.	$\geq 9/10$ successful runs, with no unrecoverable failure.	Descriptive statistics.
O1.2	Frame drop rate / jitter / CRC failure	3 long-duration runs, each lasting 50 min, under nominal load.	Frame drop $< 1\%$, p95 inter-frame time $< 1.5 \times$ nominal period, CRC failure rate = 0.	Paired t-test or Wilcoxon.
O1.3	Inter-modal alignment error	30 shared physical events, repeated across 3 runs, using controlled tapping or press-release actions.	Median FSR-accelerometer lag < 5 ms, with no systematic drift.	Wilcoxon or paired t-test.
O1.4	Host CPU usage / logging stability	3 logging runs, each lasting 50 min, under nominal load and replayed higher traffic load.	Mean CPU usage $< 70\%$, p95 CPU usage $< 85\%$, no dropped-write events.	Repeated-measures ANOVA or Friedman test.

Table A7: Evaluation metrics for O2 power scalability.

Obj.	Metric	Protocol	Acceptance criterion	Statistical test
O2.1	Stable power delivery	3 continuous runs of 30 min under full five-module operation, with rail voltage and current logged throughout.	No brownout or reset. Rail variation within $\pm 5\%$ of nominal voltage.	Descriptive statistics.
O2.2	Current draw vs sampling rate	Measure current at 100, 300, 500, and 700 Hz, using 10 min runs at each condition.	Current increases approximately linearly with sampling rate, with no unstable jump between adjacent settings.	Linear regression and repeated-measures ANOVA or Friedman test.
O2.3	Current draw vs enabled modules	Measure current with 1, 2, 3, 4, and 5 modules enabled, using 10 min runs at each condition.	Current increases approximately linearly with enabled module count.	Linear regression and repeated-measures ANOVA or Friedman test.
O2.4	Dual-patch power budget suitability	Estimate total power from measured single-patch results and validate with one 20 min emulated dual-patch load test if feasible.	Projected dual-patch power remains within supply budget, with at least 20% margin.	Descriptive analysis.

Table A8: Evaluation metrics for O3 calibration scalability.

Obj.	Metric	Protocol	Acceptance criterion	Statistical test
O3.1	Calibration repeatability	Repeat the full calibration procedure 3 times on the same five-module patch, with identical loading conditions and 10 test samples per module.	Inter-run calibration variation remains below 10%.	Repeated-measures ANOVA or Friedman test.
O3.2	Synchronization calibration error	Apply 30 shared physical events across 3 runs and compare the calibrated temporal offset between FSR and accelerometer signals.	Median residual synchronization error remains below 5 ms.	Wilcoxon signed-rank test or paired t-test.
O3.3	Inter-module variation	Apply the same reference load to all 5 modules and compare normalized output before and after calibration.	Post-calibration inter-module variation remains below 10%.	One-way ANOVA or Kruskal–Wallis test.
O3.4	Post-reassembly performance retention	Disassemble and reassemble the patch 3 times, recalibrate, and evaluate force-regression performance using the same test protocol as H1.	Post-calibration performance remains within 20% of the baseline prototype.	Paired t-test or Wilcoxon signed-rank test.

Table A9: Evaluation metrics for O4 reference design package.

Obj.	Metric	Protocol	Acceptance criterion	Statistical test
O4.1	Architecture package completeness	Review whether the electronics architecture, firmware workflow, and calibration procedure are all documented in one package.	All three components are documented and cross-referenced in a reusable form.	Checklist-based review.
O4.2	Reproducibility of design documentation	A second reader follows the documentation to trace the module design, patch integration steps, and validation workflow.	$\geq 90\%$ of required steps can be reproduced without additional clarification.	Descriptive analysis.
O4.3	Coverage of validation results	Check whether the package includes summarized results for data, power, and calibration validation.	Results from O1, O2, and O3 are all included and clearly linked to the corresponding objectives.	Checklist-based review.
O4.4	Toolkit usability for future iterations	Review whether the final package includes reusable design files, procedures, and parameter records for later patch or multi-patch extension.	The toolkit supports at least one future single-patch iteration and provides sufficient information for later multi-patch extension.	Descriptive analysis.

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